

Project Proposal. Group 7.

1. Problem Statement

Raw bicycles counts by themselves only show where the traffic volume is high, and from that alone, only so much inference can be made. They do not show, for instance, how robust the demand is to inconvenient weather conditions. Two cites could have similar average counts yet react very differently to rain: one may retain most of its cyclists, the other might lose in demand significantly. Moreover, different levels of resilience to bad weather conditions are likely not random but related to different meta-characteristics, such as the network importance (whether a certain site is used for commuting or not), route reliability, etc.

In this project, rainfall is treated as the main stress factor. The analysis therefore has two goals: first, to estimate the effect of rainfall on bicycle counts, controlling for other weather conditions such as wind, temperature, and for temporal patterns such as weekends, holidays, etc. The second goal is to explain differences in weather resilience across sites by identifying common profiles and characteristics in data associated with smaller or bigger weather-related demand drops. With this, we aim to produce a helpful and efficient characterization of different cycling locations.

2. Data Description

(https://opendata.meteo.be/geonetwork/srv/dut/catalog.search#/metadata/RMI_DATASET_SYNOP) other interesting dataset with daily weather overview

The primary dataset provided by the Agentschap Wegen en Verkeer (AWV) consists of three relational components recording traffic flow across Flanders. The core observation data is stored in monthly files containing the traffic mode, such as cyclists or pedestrians, and the count volume for specific time intervals from August 2019 to March 2026. This is supported by two additional files: sites, which provides the geographic coordinates and recording frequency for each station, and richtingen, which defines the directional orientation of the traffic. The site ID acts as the relational bridge to join traffic volumes with their spatial metadata.

Additionally the analysis will include the use of a dataset of the Royal Meteorological Institute (RMI) that includes weather observations from 17 weather stations across Belgium that report various weather related metrics on a 10 minute basis (https://opendata.meteo.be/geonetwork/srv/dut/catalog.search#/metadata/RMI_DATASET_AWS_10MIN). This dataset will be matched on the dataset of the AWV based on the reporting time. The dataset from the RMI includes 21 features of which a description can be found in the table below. The features in the RMI dataset can be broken down into 5 categories: meta- and geospatial data, atmospheric conditions, wind dynamics, soil and subsurface temperatures and solar radiation.

Feature	Type	Unit	Description
FID	String	-	Unique identifier.
the_geom	String	Degrees	Geographic coordinates in WKT (Well-Known Text) POINT format.

code	Integer	-	Unique station identification code.
timestamp	Datetime	ISO 8601	The date and time (10 min interval) of the observation.
precip_quantity	Float	mm	Total precipitation measured during the previous time interval.
temp_dry_shelter_avg	Float	°C	Average air temperature measured in a radiation shelter.
temp_grass_pt100_avg	Float	°C	Average temperature measured at grass level (0 cm).
temp_soil_avg	Float	°C	Average soil temperature (surface).
temp_soil_avg_Xcm	Float	°C	Soil temperature at depths of 5 cm, 10 cm, 20 cm, and 50 cm.
wind_speed_10m	Float	m/s	Average wind speed at a height of 10 metres.
wind_speed_avg_30m	Float	m/s	Average wind speed at a height of 30 metres.
wind_gusts_speed	Float	m/s	Maximum wind speed (gust) during the time interval.
humidity_rel_shelter_avg	Float	%	Average relative humidity measured in the shelter.
pressure	Float	hPa	Atmospheric pressure at station level.
sun_duration	Float	min/h	Duration of sunshine recorded during the time interval.
short_wave_from_sky_avg	Float	W/m ²	Average incoming short-wave solar radiation.
sun_int_avg	Float	W/m ²	Average solar intensity measured during the time interval.
qc_flags	JSON	-	Quality control validation flags for each measurement.

The data contained in this dataset span from 1995 up until today but only data from the period of August 2019 until March 2026 will be used, since the AWW dataset spans this period. The location of the closest weather station to a site in the AWW dataset will be used to attribute a weather condition to a site. Because the AWW data and RMI data have slight offsets in reporting times, rounding to the nearest timestamp is required to ensure consistent alignment across the data.

3. Methodology

Three modelling approaches will be implemented:

1. Linear Regression with Regularization (Lasso)

We will use Linear Regression to establish a baseline for how each weather feature (precipitation, wind, temperature) affects cycling counts. This provides clear coefficients (e.g., the "cost" of 1mm of rain) to quantify the direct impact of weather. Lasso can shrink irrelevant variables to zero.

2. Random Forest

We will implement a Random Forest to capture non-linear relationships and "interaction effects" (e.g., how rain affects weekend traffic differently than weekday traffic). This model is robust to outliers and can handle complex dependencies between weather and traffic patterns.

3. Clustering

Using K-Means, we will cluster sites based on their "sensitivity coefficients" derived from the regression models. This will allow us to group sites into functional profiles, such as "weather-resilient" and "weather-sensitive" sites.

The project will be implemented in Python, using the following libraries:

- **pandas** for data manipulation
- **NumPy** for numerical operations
- **scikit-learn** for preprocessing, modeling, pipelines, and clustering
- **matplotlib** and **seaborn** for visualization
- **geopandas** for matching between sites and weather stations

4. Expected Results

We expect to quantify the effect of weather conditions, in particular rainfall, on cycling traffic across different sites. The linear regression model will yield coefficients, interpretations of which will indicate how much cycling volume changes in response to variations in rainfall, temperature and wind speed. This will enable us to estimate the direct impact of adverse weather conditions on cycling traffic. We expect to find that adverse weather conditions reduce cycling traffic, with varying magnitude of this effect depending on the sites.

With the Random Forest model we expect to capture more complex and non-linear relationships, such as interaction effects between weather variables and temporal factors— for example differences between weekdays and weekends. Through this approach, the predictive performance is expected to improve, and patterns that are not identified by the linear model can be uncovered, such as differing sensitivities depending on time and context.

We expect to identify clusters of cycling sites, with the usage of the sensitivity measures derived from the aforementioned analyses. These clusters are likely to reflect meaningful differences, such as weather-resilient sites where cycling traffic remains relatively stable despite rainfall, and weather-sensitive sites where traffic decreases significantly under adverse weather conditions.

It is our expectation that the findings of this research will yield meaningful information about the effects of weather conditions on cycling behaviour. These findings can be particularly useful to agencies such as AWW for the purposes of monitoring cycling traffic trends and infrastructure usage, evaluating the impact of new cycling paths or road changes, planning maintenance and improvements efficiently, and making informed policy decisions aimed at promoting cycling as a sustainable mode of transport.